

Sample p1

1) a)

$$P(ML) = 0.45$$

$$P(ISM) = 0.6$$

$$P(DL) = 0.5$$

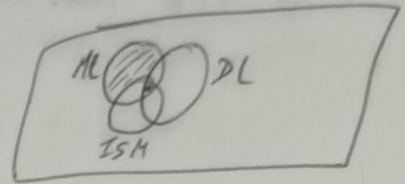
$$P(ML \& ISM) = 0.15$$

$$P(ISM \& DL) = 0.25$$

$$P(ML \& DL) = 0.2$$

$$P(ML \& DL \& ISM) = 0.05$$

$$P(ML \text{ or } DL \text{ or } ISM) = 1$$



$\therefore$ Data is valid

$$\begin{aligned} \text{i) } P(\text{only ML}) &= P(ML) - P(ML \& ISM) - P(ML \& DL) + P(ML \& ISM \& DL) \\ &= \boxed{0.15} \end{aligned}$$

$$\begin{aligned} \text{ii) } P(ML \& \text{not } DL) &= P(ML) - P(ML \& DL) \\ &= 0.45 - 0.2 \\ &= \boxed{0.25} \end{aligned}$$

$$\begin{aligned} \text{iii) } P(ML \& ISM \& \text{not } DL) &= P(ML \& ISM) - P(ML \& ISM \& DL) \\ &= 0.15 - 0.05 \\ &= \boxed{0.1} \end{aligned}$$

$$\begin{aligned} \text{iv) } P(ML \& DL \& \text{not } ISM) &= P(ML \& DL) - P(ML \& DL \& ISM) \\ &= 0.2 - 0.05 \\ &= \boxed{0.15} \end{aligned}$$

b)

$$p = 0.6 \quad q = 0.4$$

$$n = 120$$

X = choosing a random smoker & he uses that brand

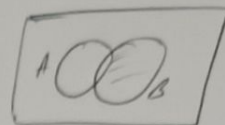
To find: $P(X \leq 60)$ or $P(X \leq 60)$

$$P(X \leq 60) = \sum_{i=0}^{60} {}^{120}C_i p^i q^{120-i}$$

$$= 0.0167$$

$$= 1.67\%$$

$$27) a) P(A) = 0.4 \quad P(B) = 0.3 \quad P(A \cap B) = 0.2$$



$$P(A \cup B) = 0.4 + 0.3 - 0.2 = 0.5$$

$$i) P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{0.4}{0.5} = 0.8$$

$$ii) P(A|A \cap B) = \frac{P(A \cap (A \cap B))}{P(A \cap B)} = 1$$

$$iii) P(A'|A \cup B) = \frac{P(A' \cap (A \cup B))}{P(A \cup B)} = \frac{P(B) - P(A \cap B)}{P(A \cup B)} = \frac{0.1}{0.5} = 0.2$$

$$iv) P(A'|A \cap B) = \frac{P(A' \cap (A \cap B))}{P(A \cap B)} = 0$$

$$b) \begin{array}{l} P(S) = 0.6 \Rightarrow P(S') = 0.4 \\ P(V) = 0.4 \Rightarrow P(V') = 0.6 \end{array} \quad \left| \begin{array}{l} P(P|S) = 0.15 \Rightarrow P(P|S') = 0 \\ P(P|V) = 0.25 \Rightarrow P(P|V') = 0 \end{array} \right.$$

$$i) P(S'|P) = \frac{P(P|S')P(S')}{P(P)} = 0$$

$$ii) P(V'|P) = \frac{P(P|V')P(V')}{P(P)} = 0$$

$$iii) P(S'|P)P(V'|P) = 0$$

$$3) a) f(x, y) = \frac{x+ky}{2} \quad \begin{array}{l} 0 < x < 1 \\ 0 < y < 2 \end{array}$$

i) conditions for a f^1 to be PDF:

• non-negative :- $f(x, y) \geq 0 \quad \forall x, y$

$\Rightarrow x > 0$ & $y > 0$, it's possible for $f(x, y) \geq 0$.

• Total area = 1 :-

$$\iint_{x, y} f(x, y) dx dy = 1$$

for valid values of k , $f(x, y)$ can be a PDF.

• independence :- $P(X=x, Y=y) = 0$,

$$\text{for valid values of } k, \iint_{x, y} f(x, y) dx dy = 0.$$

$\therefore f(x, y)$ can be a PDF.

ii) Find k .

$$\int_0^1 \int_0^2 \frac{x+ky}{2} dx dy = 1$$

$$\Rightarrow \int_0^2 \left[\frac{x^2}{2} + \frac{kxy}{2} \right]_0^2 dy = 1$$

$$\Rightarrow \int_0^2 \left(\frac{1}{2} + \frac{ky}{2} \right) dy = 1$$

$$\Rightarrow \left[\frac{y}{2} + \frac{ky^2}{4} \right]_0^2 = 1$$

$$\Rightarrow \frac{1}{2} + k = 1$$

$$\Rightarrow k = \frac{1}{2}$$

Mean

$$\int_0^1 \int_0^2 xy \left(\frac{x + \frac{1}{2}y}{2} \right) dx dy$$

Variance

$$\int_0^1 \int_0^2 x^2 y^2 \left(\frac{x + \frac{1}{2}y}{2} \right) dx dy$$

3) b)

i) $0.5 = 1-1$

y	x		
	0	1	2
-1	0.05	0.125	0.075
0	0.07	0.175	0.105
1	0.08	0.2	0.12

$$P(x=x, y=y) = P(x=x) * P(y=y)$$

$$\sum P(x=x, y=y) = 1$$

$$i) P(x < 2 | y = -1) = \frac{P(x < 2, y = -1)}{P(y = -1)}$$

$$= \frac{0.05 + 0.125}{0.05 + 0.125 + 0.075}$$

$$= \boxed{0.7}$$

$$ii) P(x < 1 | y < 1) = \frac{P(x < 1, y < 1)}{P(y < 1)}$$

$$= \frac{0.05 + 0.07}{0.05 + 0.125 + 0.075 + 0.07 + 0.175 + 0.105} \text{ or } \frac{0.05 + 0.07}{1 - 0.4}$$

$$= \boxed{0.2}$$

9) a)

$$\mu = 65$$

$$\sigma = 5$$

$$n = 45$$

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{45}} = 0.745$$

$$i) z_1 = \frac{55 - \mu}{SE} = -13.423$$

$$P(z_1 < z < z_2) \approx 1$$

$$z_2 = \frac{70 - \mu}{SE} = 6.711$$

$\therefore z_1 < SE < z_2 \therefore$ mean lies in 55 & 70

$$ii) \bar{x} = 69.99$$

$$z_1 = 6.698$$

$$\bar{x}_2 = 70.01$$

$$z_2 = 6.725$$

$$P(z_1 < z < z_2) = P(z < z_2) - P(z < z_1)$$

$$= 1 - 1 = 0$$

$$b) \mu = 20$$

$$\sigma^2 = 4 \Rightarrow \sigma = 2$$

normal distribution

$$i) P(X > 25) = 1 - P\left(z < \frac{25 - 20}{2}\right)$$

$$= 1 - 0.894 = 0.994$$

$$= 0.006$$

$$ii) P(15 < X < 28) = P\left(z < \frac{28 - 20}{2}\right) - P\left(z < \frac{15 - 20}{2}\right)$$

$$= 0.999 - 0.006$$

$$= 0.993$$

$$iii) P(X = 22) = 0$$